

Session Handoff — Code Review Fixes

DCAK RTA v8 · Confidence-Estimation Pipeline (DavidsDatasets/) · 2026-07-06 → 07

Context: A full review of the pipeline found 20 issues; all were fixed, tested, and left **uncommitted** for review. Nothing changes prompts or decoding behavior — all 54 prompt strings are byte-pinned and proven identical to the pre-refactor code.

Result-corrupting bugs P0 — old CSVs affected

#	Problem	Fix
1	<code>seq_confidence_mean</code> was actually the log-prob sum → AUROC partly measured answer length, not confidence	Now a true per-token mean; sum kept as new <code>seq_log_prob_sum</code> column; both shown in the AUROC table. ⚠ In old CSVs the column still holds the sum.
2	Calibration/ECE binned 1–10 values on a 0–1 grid → every row became NaN, ECE was never computed	Rubric-aware binning (class N → midpoint (N–0.5)/10); ECE vs empirical bin confidence. Verified on the real MedQA CSV.
3	<code>verbalized_confidence</code> silently mixed two-pass and single-pass scores on difficulty-correlated rows	New <code>two_pass_confidence</code> column + <code>verbalized_conf_source</code> tracking; per-method AUROC & calibration reported separately.
4	Failed samples were silently dropped → accuracy biased toward easy items	Failures logged with traceback, loud bias warning with failure rate, saved to <code>*_errors_*.json</code> sidecar.
5	Config claimed SE enabled while debug flags silently disabled it (NLI model still loaded onto GPU)	Single <code>SEMANTIC_ENTROPY_ACTIVE</code> flag consumed everywhere; loud import-time warning when vetoed.
6	Only numpy was seeded → sampled generation non-reproducible despite the advertised seed	<code>set_seed()</code> seeds python / numpy / torch CPU+CUDA at both entry points.

Correctness & robustness P1

#	Problem	Fix
7	Orphaned SE helper crashed with <code>TypeError</code> if ever called	Removed (zero callers confirmed, incl. notebooks).
8	Base/GPT-OSS rows decoded under different anti-loop guard constraints — an invisible cross-family confound	New <code>decoding_guards_active</code> column per row so analyses can condition on decoding policy.
9	TriviaQA substring match: "no" c "North Carolina", "art" c "Descartes" scored as correct	Word-boundary containment + length floor + stopword block; 16-case regression suite passes.
10	Answer-letter entropy discarded the pre-renormalization probability mass (its reliability signal)	Persisted as <code>answer_letter_mass</code> column — low mass ⇒ that row's entropy is untrustworthy.

Performance & infrastructure P1

#	Problem	Fix
11	Serial batch-size-1 generation (up to 5 sequential <code>generate</code> calls per sample)	Opt-in <code>DCAK_GEN1_BATCH_SIZE</code> batched Gen-1 (guarded families auto-fallback to serial); validate per model with <code>smoke_test_batched.py</code> before enabling.
12	Guarded clean re-scoring pinned multi-GB logit tensors on GPU per sample	Only needed positions materialized (<code>num_logits_to_keep</code> + version fallbacks), slice cloned, offloaded to CPU.

13	Crash at row N lost every completed result; CSV round-trip mangled Unicode (the "â" artifacts)	<code>IncrementalJSONLWriter</code> : per-row flushed <code>*_rows_*.jsonl</code> , exact Unicode round-trip — the new source of truth.
14	Smoke-test row evaluated twice; detected device never passed to the loader (CPU-only crashed)	Smoke result cached & reused; device passed through with CPU fallback.
15	<code>trust_remote_code=True</code> everywhere with no revision pinning (supply-chain + reproducibility hole)	All 12 models + 6 datasets + NLI + PRM pinned to HF commit SHAs; flag dropped for natively-supported families.
16	Any config change required editing source (blocked sweeps)	<code>DCAK_*</code> env overrides for every knob, e.g. <code>DCAK_DATASET=gsm8k python3 main.py</code> ; derived values track automatically.

Maintainability & hygiene P2

#	Problem	Fix
17	Rubric x3, harmony delimiter x3, extraction helpers x2 copy-pasted across modules	Single-sourced in new <code>shared.py</code> ; Gen-2/3 prompts extracted as pure builders — proven byte-identical to git HEAD .
18	Regex extractors untestable — every model added fragile special cases	<code>check_extraction_golden.py</code> pins behavior over 1,410 real captured responses ; <code>check_prompt_golden.py</code> pins all 54 prompts.
19	<code>prm_scoring.py</code> loaded a 7B model on import, hardcoded paths, no main guard	Proper CLI with <code>main()</code> guard, safe token-id lookup, added the AUROC/std the TODO asked for.
20	116 junk files tracked (checkpoint copies, <code>.pyc</code> , <code>.DS_Store</code> , 0-byte <code>=*</code>); blanket warning filter muted the pipeline's own diagnostics	<code>.gitignore</code> added, junk staged for deletion, warning filter narrowed, <code>plt.close()</code> added, <code>requirements.txt</code> created.

Verification — all green, no GPU needed

`verify_rubric.py` — ALL CHECKS PASSED · `check_prompt_golden.py` — 54/54 byte-identical · `check_extraction_golden.py` — 1,410/1,410 no drift · `tests/` — 3 mock suites pass (batched ≡ serial decoding, guarded-path re-scoring, full `evaluate_sample` e2e incl. merge + NaN policy). See `tests/README.md`.

Your next steps

- 1. Review `git status` → commit (114 staged deletions + 11 modified + 9 new files).
- 2. On the GPU box: run `python3 smoke_test_batched.py` once per model family before enabling `DCAK_GEN1_BATCH_SIZE>1`; run `pip freeze` to pin exact versions into `requirements.txt`.
- 3. Remember: **old CSVs** ' `seq_confidence_mean` holds the sum; `Trash/` was left for you to delete.
- 4. For real SE runs: set `SKIP_NLI_CLUSTERING=False` and `SE_NUM_SAMPLES=5`.